Looking at Markov Chain Convergence in General

Viren Kumar

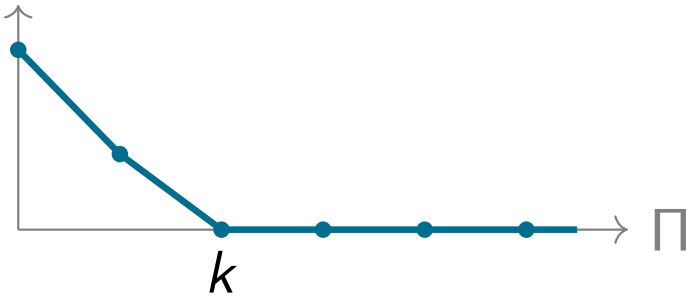
Dr. Alan Krinik

Dr. Randall Swift

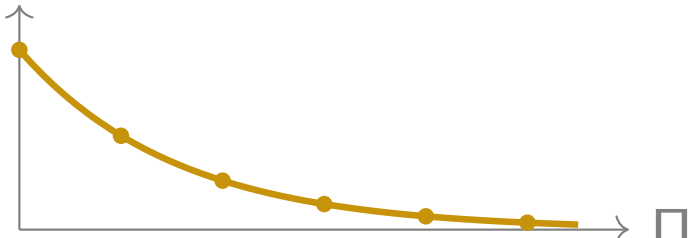
AMS Fall 2026 Western Sectional Meeting

Conjecture

A n size markov chain with a steady state can reach its steady state only within n-1 steps

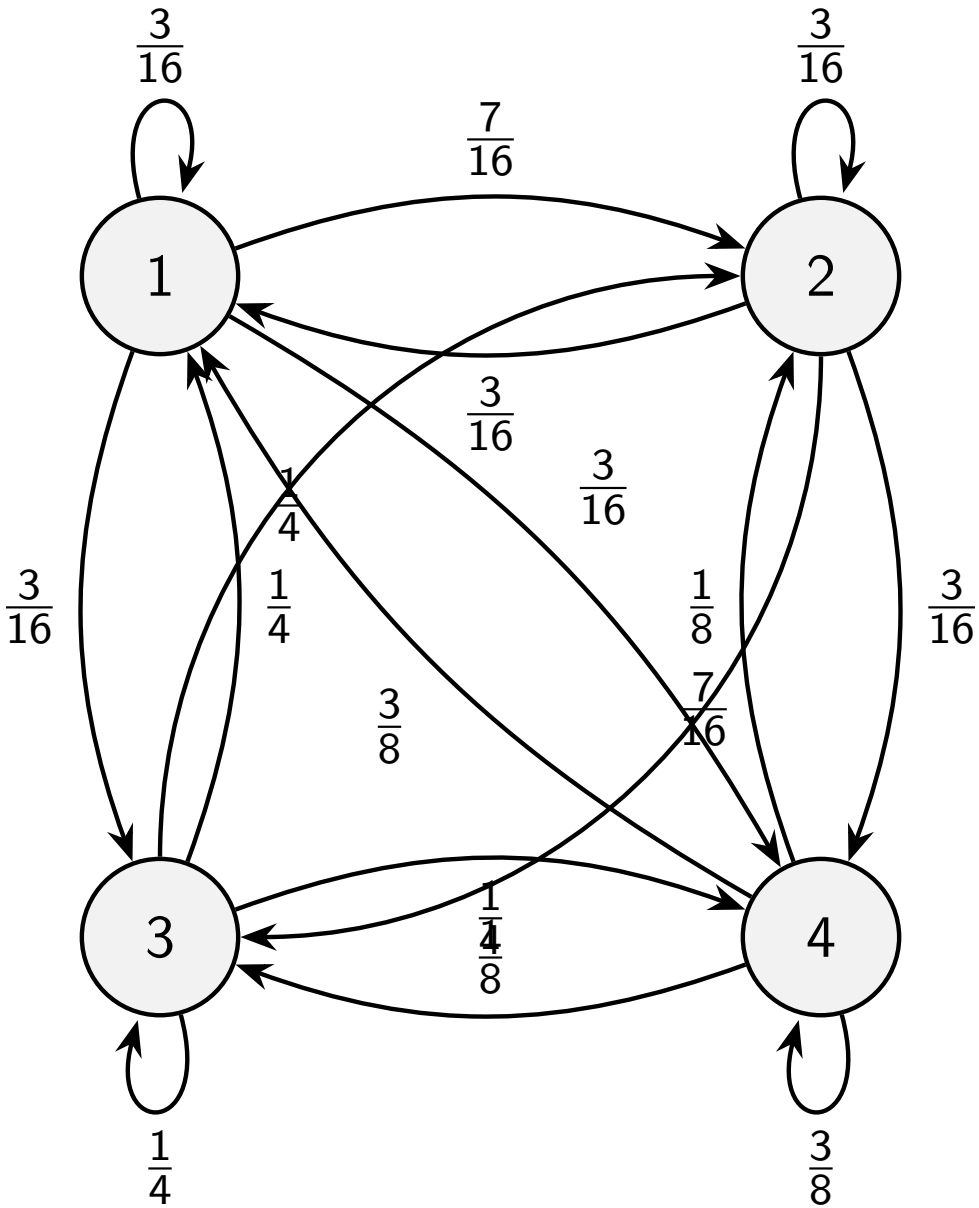


$$P^k = \Pi$$



$$P^m \longrightarrow \Pi$$

A 4 state example



A 4 State Example

$$P = \begin{pmatrix} \frac{3}{16} & \frac{7}{16} & \frac{3}{16} & \frac{3}{16} \\ \frac{3}{16} & \frac{3}{16} & \frac{7}{16} & \frac{3}{16} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{3}{8} & \frac{1}{8} & \frac{1}{8} & \frac{3}{8} \end{pmatrix}$$

$$P^2 = \begin{pmatrix} \frac{15}{64} & \frac{15}{64} & \frac{19}{64} & \frac{15}{64} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{17}{64} & \frac{17}{64} & \frac{13}{64} & \frac{17}{64} \end{pmatrix}$$

$$P^3 = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix} = \Pi$$

$$\pi = \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$$

An n-1 steady state generator

Construction: For $r \geq 3$, define P_r by

$$(P_n)_{ij} = \begin{cases} \dfrac{2n-1}{n^2}, & 1 \leq i \leq n-2,\ j = i+1, \\ \dfrac{n-1}{n^2}, & 1 \leq i \leq n-2,\ j \neq i+1, \\ \dfrac{1}{n}, & i = n-1, \\ \dfrac{2(n-1)}{n^2}, & i = n,\ j \in \{1, n\}, \\ \dfrac{n-2}{n^2}, & i = n,\ 2 \leq j \leq n-1. \end{cases} \tag{1}$$

Here

$$P_n^{n-2} \neq J_n, \qquad P_n^{n-1} = J_n.$$

Every entry of P_n is strictly positive. The matrix is doubly stochastic, so its invariant distribution is uniform. Positivity makes the chain irreducible and aperiodic. Hence the first steady-state power is exactly P_r^{n-1}.

Conjecture

A markov chain that reaches its steady state has one eigenvalue of 1, with the rest being zero.

Distinct eigenvalues: Sylvester's expansion

With $\lambda_1 = 1$, $A_1 = \Pi$, and $|\lambda_i| < 1$ for $i \geq 2$,

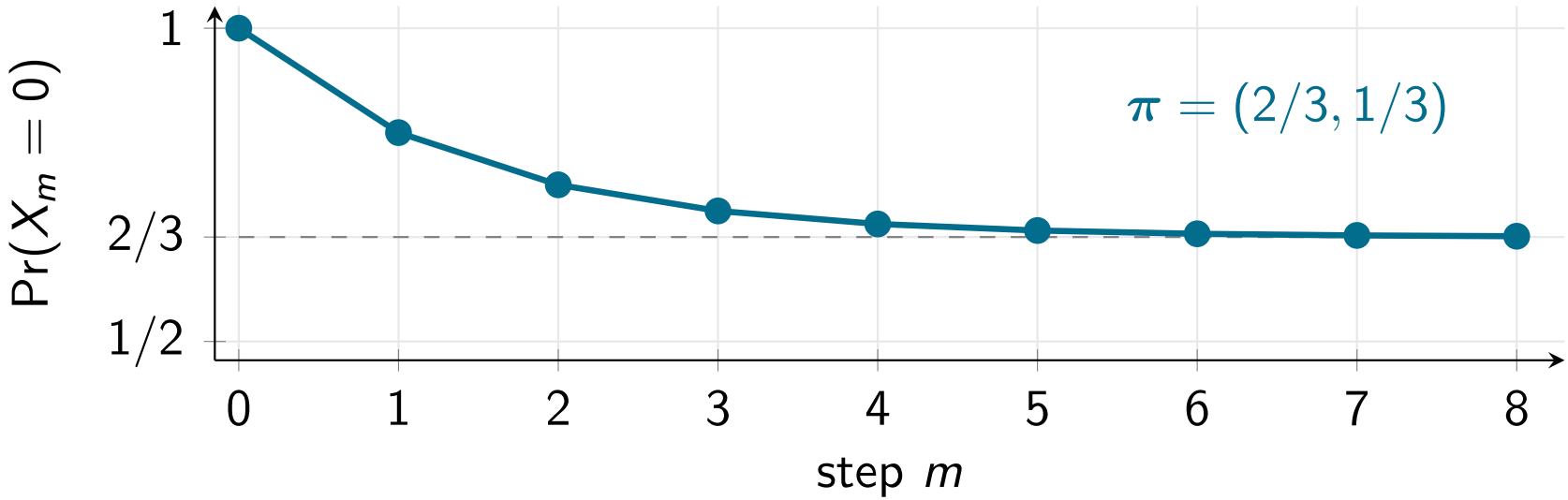
$$P^m = \sum_{i=1}^{n} \lambda_i^m A_i = \Pi + \sum_{i=2}^{n} \lambda_i^m A_i.$$

Here A_i is the spectral projector for λ_i.

Example $\lambda_1 = 1, \lambda_i = \frac{1}{2}$

$$P_{1/2} = \begin{pmatrix} 5/6 & 1/6 \\ 1/3 & 2/3 \end{pmatrix}, \qquad \mu_0 = (1, 0).$$

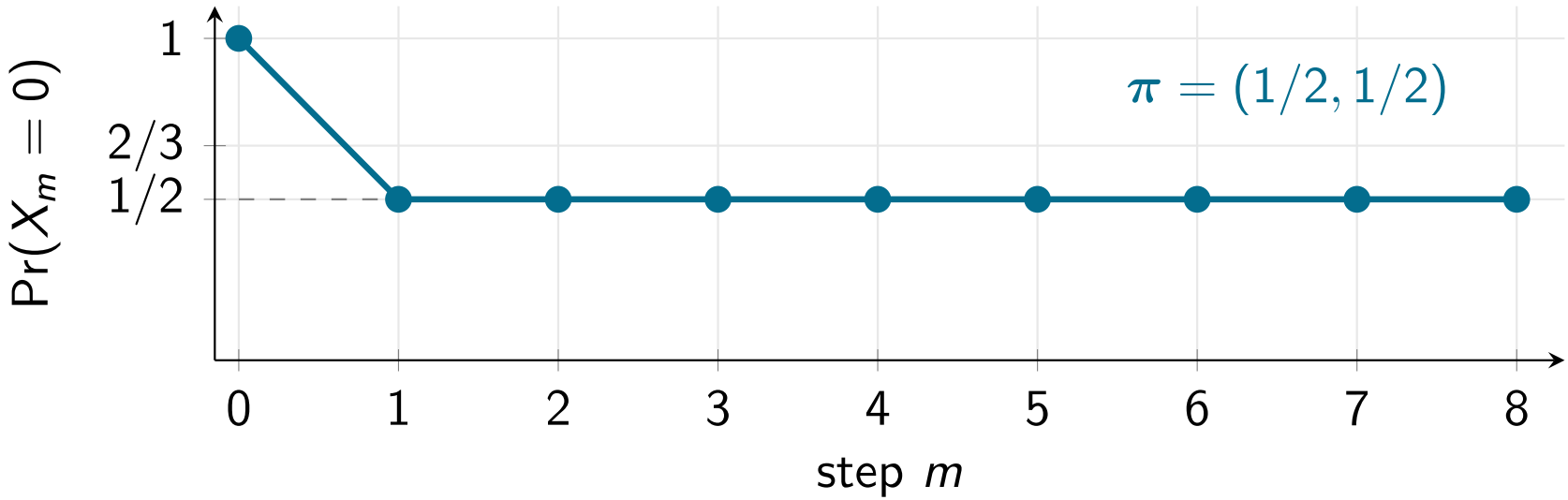
$$\Pr(X_m = 0) = \frac{2}{3} + \frac{1}{3}\left(\frac{1}{2}\right)^m.$$



Example: $\lambda_1 = 1, \lambda_i = 0$

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \qquad \mu_0 = (1, 0).$$

$$\Pr(X_m = 0) = \frac{1}{2} + \frac{1}{2}\delta_{m,0}.$$



An attempt at a Generalization

Let P be some markov chain with a with a finite step steady state, $m \in \mathbb{N}$

$$P^m = \Pi + N^m$$

where at some finite-step k

$$P^k = \Pi$$

Since the chain reaches its steady state at some finite step k, $P^k = \Pi$, which implies $N^k = 0$. Thus, N is a nilpotent matrix.

By Jordan Canonical Form, the maximum nilpotent block size in an $(n-1)$-dimensional subspace is $(n-1) \times (n-1)$.

Therefore, absolute finite-step convergence can only occur when $k \leq n-1$.

References



Douglas W. Rumble.

Jordan Form and Nilpotent Matrices.

Millersville University Department of Mathematics.

Available online:
https://sites.millersville.edu/rumble/Math.422/jordannil.pdf



Hoel, P. G., Port, S. C., & Stone, C. J. (1972).

Introduction to Stochastic Processes.

Chapter 4.1: Mean and Covariance Functions, pp. 110–119.



Meyer, C. D. (2023).

Matrix Analysis and Applied Linear Algebra.

Society for Industrial and Applied Mathematics.



Parzen, E. (1967).

Stochastic Processes.

Random Variables and Stochastic Processes, pp. 7–40.

https://doi.org/10.1137/1.9781611971120.ch1.ch6